

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2023 - 2024)

JEE(Advanced)**PRE-SCORE****22-10-2023****JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)**

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 186

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

READ THE INSTRUCTIONS CAREFULLY**GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

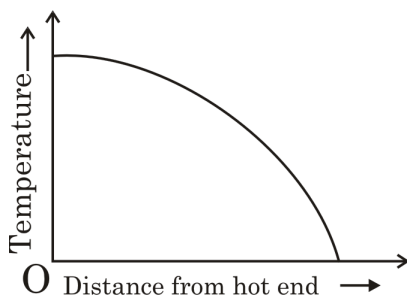
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. The tips of two identical syringes were connected using a short rubber tube. Initially, the volume of air in each of the syringes was $V = 8$ ml, the pressure inside system was equal to atmospheric pressure = 100 kPa. Cross-sectional area syringe is $S = 3 \text{ cm}^2$. Holding the syringe bodies motionless, the piston of syringe No. 1 is pressed so that it begins to move very slowly with a constant speed. On the piston of syringe No. 2, no additional force is applied. At the point in time when the volume of air in syringe No. 1 was $V_1 = 6$ ml, the volume of air in syringe No. 2 was $V_2 = 9$ ml. There is no stop anywhere in the syringes.

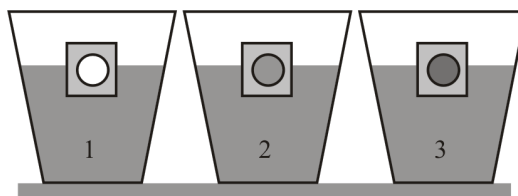


- (A) The piston of syringe 2 remained motionless for some time and then started moving
- (B) The piston of syringe 2 moved initially and then abruptly stopped
- (C) The pressure in the syringes in the final position is approximately 106.6 kPa
- (D) The pressure in the syringes in the final position is approximately 110.2 kPa

2. The ends of a metallic bar are maintained at different temperature. In the steady state the temperature variation along the length of the bar is as shown in the figure. This could possibly be due to :

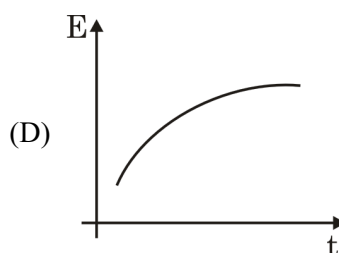
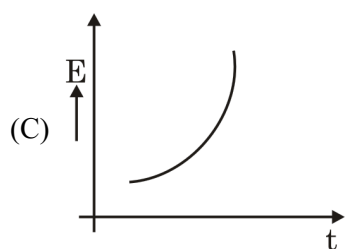
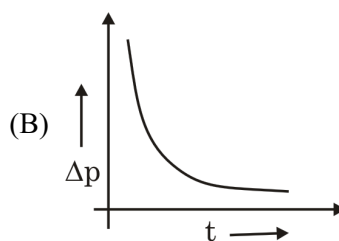
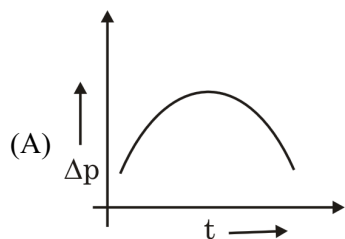


- (A) No heat loss to the surroundings along the length of the bar. The area of cross-section is decreasing as we move from hot end to cold end.
- (B) No heat loss to the surroundings along the length of the bar. The coefficient of thermal conductivity is decreasing as we moved from hot end to cold end.
- (C) No heat loss to the surroundings along the length of the bar. the area of cross-section is increasing as we move from hot end to cold end.
- (D) The bar is uniform but heat is being lost from the sides along the length of the bar to the surroundings.
3. A cubical piece of metal floats on mercury (Density of metal is half the density of mercury) such that its flat face is horizontal. The coefficient of volume expansion of metal and mercury are γ and 2γ respectively. If the temperature of both mercury and metal are increased by an amount ΔT ,
- (A) The fraction of volume inside the mercury increases by $2\gamma \Delta T$.
- (B) The Buoyant force remains the same.
- (C) The fraction of height of the cube above mercury increases by $\gamma \Delta T$.
- (D) The area of the cube increases by a factor of $(1 + \frac{2\gamma \Delta T}{3})$
4. Three ice cubes are floating in each glass containing water. The ice cube in glass 1 has an air bubble in it. The ice cube in glass 2 has a cavity inside filled with water and the third ice cube floats in glass 3 with an aluminium core. What can be said about the water level in the glasses after ice has melted ?

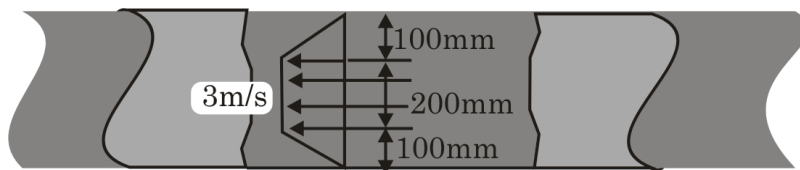


- (A) The water in glass 1 has risen.
- (B) The water level in glass 3 has got down.
- (C) The water level in glass 2 is unchanged.
- (D) The water level in all glass is unchanged.

5. A soap bubble is very slowly blown at the end of a glass tube by a pump which supplies a fixed volume of air every minute. Here we draw the graphs of excess pressure p with time t , surface energy E with time t , choose the correct option(s)

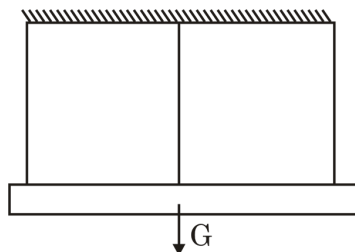


6. The velocity profile of a liquid flowing through the pipe is approximated by the truncated conical distribution as shown. The liquid has a density of $\rho = 900 \text{ kg/m}^3$. The coefficient of viscosity is 0.01 Ns/m . Assume Newton's law.



- (A) The shear stress at the walls is same as the shear stress at a distance of 150 mm from the axis.
- (B) The shear stress at the axis is zero.
- (C) The force of fluid friction acting on the walls per unit length is $\frac{12\pi}{100} \text{ N/m}$.
- (D) The rate of volume flow through the pipe is $\frac{7\pi}{100} \text{ m}^3/\text{s}$.

7. A partly full water tank admits water at 20°C and 85 N/s weight flow while ejecting water on the other side at $5500 \text{ cm}^3/\text{s}$. The air picket in the tank has a vent at the top and is at 20°C and 1 atm . If the fluids are approximately in-compressible, how much air is passing through the vent ? In which direction ? Density of water = 1000 kg/m^3 , $g = 10 \text{ m/s}^2$.
- (A) Air is flowing in
- (B) Air is flowing out.
- (C) $\left| \frac{dV}{dt} \right| = 3000 \text{ cm}^3/\text{s}$
- (D) $\left| \frac{dV}{dt} \right| = 4650 \text{ cm}^3/\text{s}$
8. A homogeneous block with a mass $m = 100 \text{ kg}$ hangs on three light vertical wires of equal natural lengths arranged symmetrically (Figure). Find the tension in the wires if the middle wire is of steel and the other two are of copper. All the wires have the same cross section. Consider the modulus of elasticity of steel to be double that of copper. ($g = 10 \text{ m/s}^2$)



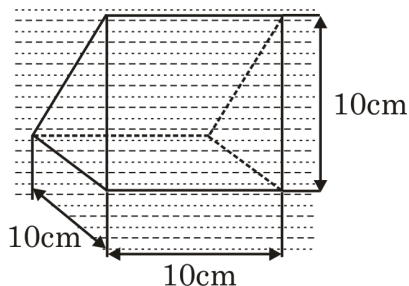
- (A) Tension in steel wire = 250 N
- (B) Tension in copper wire = 500 N
- (C) Tension in steel wire = 500 N
- (D) Tension in copper wire = 250 N

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

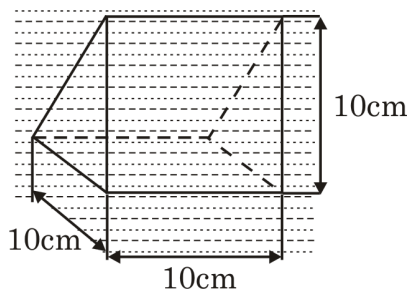
9. A solid right angled prism of dimensions shown here is submerged into a liquid such that its base is horizontal. The density of the liquid is 1gm/cc and the density of the prism is ρ . List-I gives the details of the face of the prism and List-II gives the force exerted by the liquid on it. Assume that the top vertex of the prism is at the surface of liquid. Neglect atmospheric pressure.



	List-I		List-II
(I)	Inclined face	(P)	10 N
(II)	vertical triangular face	(Q)	5 N
(III)	Base	(R)	$10/3$ N
(IV)	Flat vertical rectangular face	(S)	$5/3$ N
		(T)	$20/3$ N
		(U)	$5\sqrt{2}$ N

- (A) $I \rightarrow Q; II \rightarrow R; III \rightarrow P; IV \rightarrow S$
 (B) $I \rightarrow U; II \rightarrow S; III \rightarrow P; IV \rightarrow Q$
 (C) $I \rightarrow U; II \rightarrow R; III \rightarrow P; IV \rightarrow Q$
 (D) $I \rightarrow U; II \rightarrow T; III \rightarrow P; IV \rightarrow S$

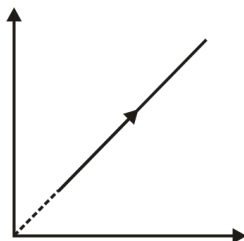
10. A solid right angled prism of dimensions shown here is submerged into a liquid such that its base is horizontal. The density of the liquid is ρ and the density of the prism is 1 gm/cc . List-I gives the details of the density ρ and List-II gives the acceleration of COM of the prism and angular acceleration of prism. Assume that the top vertex of the prism is at the surface of liquid. Neglect atmospheric pressure.



	List-I		List-II
(I)	1 gm/cc	(P)	Acceleration is zero
(II)	2 gm/cc	(Q)	Angular acceleration is zero
(III)	0.5 gm/cc from top till depth of 10 cm and 2 gm/cc below the prism	(R)	Acceleration is upwards
(IV)	1 gm/cc from top till depth of 10 cm and 2 gm/cc below the prism	(S)	Angular acceleration is clockwise
		(T)	Angular acceleration is anticlockwise
		(U)	Acceleration is downwards

- (A) $I \rightarrow P, Q; II \rightarrow Q, R; III \rightarrow Q, U; IV \rightarrow P, Q$
 (B) $I \rightarrow P, Q; II \rightarrow Q, U; III \rightarrow Q, R; IV \rightarrow P, Q$
 (C) $I \rightarrow P, Q; II \rightarrow Q, R; III \rightarrow S, U; IV \rightarrow T, Q$
 (D) $I \rightarrow P, Q; II \rightarrow Q, R; III \rightarrow T, U; IV \rightarrow T, U$

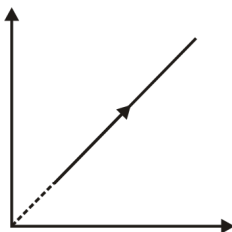
11. In the archives of Lord Kelvin, researchers discover a graph which is shown below. It refers to processes carried out on a monoatomic ideal gas in the direction shown. But it is not known that what is plotted on x axis and what on y axis. There is a detailed discussion amongst researchers as to what is the pressure change, volume change and temperature change in each of the processes. List-I gives the suggested variables for x and y axis. List-II gives the effect on the parameters shown. Match the appropriate entries.



	List-I		List-II
(I)	Temperature on x axis and pressure on y axis	(P)	Pressure is increasing
(II)	Temperature on x axis and volume on y axis	(Q)	Pressure is decreasing
(III)	Volume on x axis and pressure on y axis	(R)	Volume is increasing
(IV)	entropy on x axis and temperature on y axis	(S)	Volume is decreasing
		(T)	Temperature is increasing
		(U)	Temperature is decreasing

- (A) I $\rightarrow$ P,T; II $\rightarrow$ R,T; III $\rightarrow$ P,R,T; IV $\rightarrow$ Q,R,T
 (B) I $\rightarrow$ P,T; II $\rightarrow$ R,T; III $\rightarrow$ P,R,T; IV $\rightarrow$ T
 (C) I $\rightarrow$ P,T; II $\rightarrow$ Q,T; III $\rightarrow$ P,R,T; IV $\rightarrow$ Q,S,T
 (D) I $\rightarrow$ P,U; II $\rightarrow$ R,U; III $\rightarrow$ P,R,U; IV $\rightarrow$ Q,S,T

12. In the archives of Lord Kelvin, researchers discover a graph which is shown below. It refers to process carried out on a monoatomic ideal gas in the direction shown. But it is not known that what is plotted on x axis and what on y axis. There is a detailed discussion amongst researchers as to what is the heat given, work done by the gas and internal energy change in each of the processes. List-I gives the suggested variables for x and y axis. List-II gives the effect on the parameters shown. Match the appropriate entries.



	List-I		List-II
(I)	Temperature on x axis and pressure on y axis	(P)	Work is done by the gas
(II)	Temperature on x axis and volume on y axis	(Q)	Work is done on the gas
(III)	Volume on x axis and pressure on y axis	(R)	Heat is given to the gas
(IV)	entropy on x axis and temperature on y axis	(S)	Heat is extracted from the gas
		(T)	Internal energy of the gas increases
		(U)	Internal energy of the gas decreases

- (A) I $\rightarrow$ Q,S,U; II $\rightarrow$ Q,S,T; III $\rightarrow$ P,R,U; IV $\rightarrow$ T
 (B) I $\rightarrow$ R,T; II $\rightarrow$ Q,R,T; III $\rightarrow$ Q,R,T; IV $\rightarrow$ Q,R,T
 (C) I $\rightarrow$ R,T; II $\rightarrow$ P,R,T; III $\rightarrow$ P,R,T; IV $\rightarrow$ P,R,T
 (D) I $\rightarrow$ R,T; II $\rightarrow$ P,R,T; III $\rightarrow$ P,R,T; IV $\rightarrow$ R,T

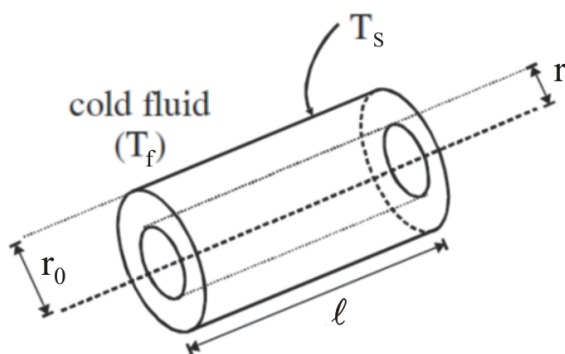
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

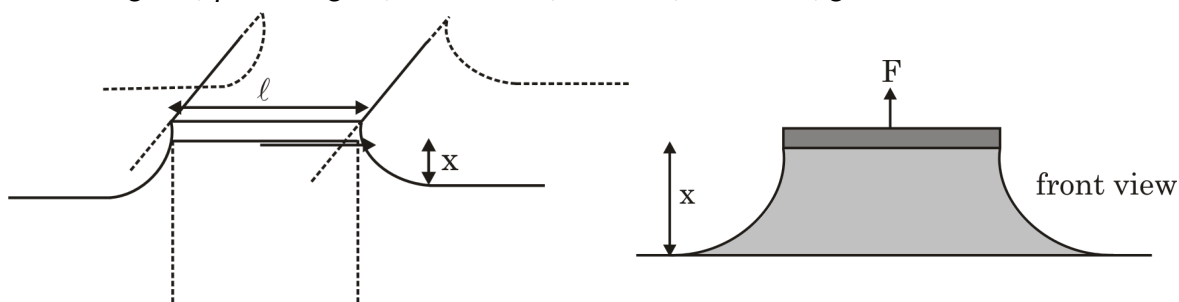
Zero Marks : 0 In all other cases.

1. The heat machine of the room which works as a split-system operates on the inverse Carnot cycle, and it can be assumed that the room and outdoor temperatures correspond to the temperatures on the isotherms of the cycle. In summer, when the temperature outside the window is $+27^\circ\text{C}$. The split system, working in air conditioning mode (as a refrigeration unit). keeps the room temperature at $+17^\circ\text{C}$ and consumes from the power grid an average power N_1 . At the beginning of winter, when the temperature outside drops to -3°C . the split system switches to heat pump mode and maintains the same room temperature of $+17^\circ\text{C}$ as in the summer, by consuming average power N_2 . We can assume that the heat flow (through the windows and walls) is proportional to the temperature difference between the room and outside, with same coefficient of proportionality in summer and winter. Find the ratio of powers $n = N_2/N_1$.
2. Consider a long, solid cylinder as shown in the figure, which could represent a current-carrying wire or a fuel element in a nuclear reactor. For steady state conditions, the rate at which heat is generated within the cylinder must equal the rate at which heat is convected from the surface of the cylinder to a moving fluid. This condition allows the surface temperature to be maintained at a fixed value of T_s . The rate of heat generated per unit volume is 20 kW/m^3 , the coefficient of thermal conductivity is 5 W/mK , what is the temperature (in K) at the distance $r = R_0/2$ from the axis. Take $R_0 = 20\text{ cm}$. The temperature T_s at the surface is 300 K .

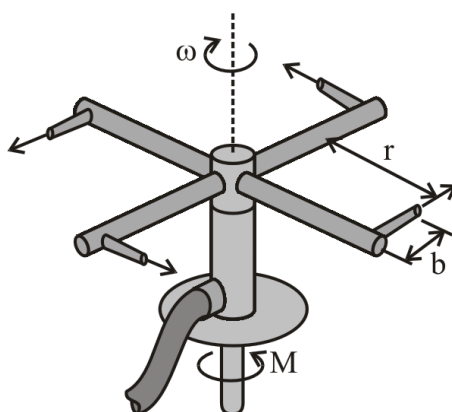


3. A liquid takes 5 minutes to cool from 80°C to 50°C . How much time (in mins.) will it take to cool from 60°C to 30°C ? The temperature of surrounding is 20°C . Assume Newton's law. (Take $\ln 2 = 0.69$, $\ln 3 = 1.10$)

4. It is known that in a pan kept on the stove, 1 liter of water is heated by 10 degrees in 42 s. The burner power is constant, heat capacity of pan is negligible, heat loss is negligible. Specific heat capacity of water 4200 J/(kg °C), specific heat capacity of ice 2100 J/(kg °C), specific heat of melting of ice 330 kJ/kg. How long does it take to heat on the same burner to boiling point of water a mixture of 1 liter of water and 3 kilograms of ice? Initially, the mixture is at the melting temperature of ice. Give your answer in minutes.
5. A long plate of width ℓ is brought into contact with the liquid surface. Then the plate is pulled up. Find the value of the force exerted by us per unit length (in N/m) of the plate. Liquid density ρ and surface tension (σ). λ = mass per unit length.
(Take $\lambda = 10 \text{ gm/m}$, $\rho = 10^3 \text{ kg/m}^3$, $\sigma = 0.5 \text{ N/m}$, $x = 2 \text{ mm}$, $\ell = 10 \text{ cm}$, $g = 10 \text{ m/s}^2$.)



6. A sprinkler has four outlets as shown. All of them have uniform cross sectional area of 10 cm^2 . Water of density 1000 kg/m^3 enters the system at the centre as shown at a flow rate of 200 cc/sec. The ejected water exerts a torque so that in the steady state, the sprinkler rotates with a constant angular velocity ω as shown. There is no external torque. Neglect friction. Find the steady state angular velocity (in rad/s) of the sprinkler. $r = 5 \text{ cm}$, $b = 1 \text{ mm}$.



PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

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Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

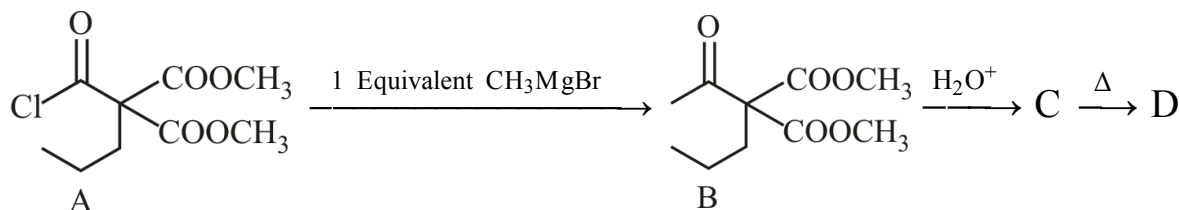
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

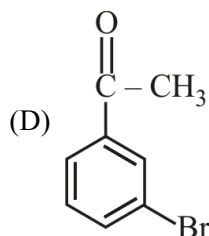
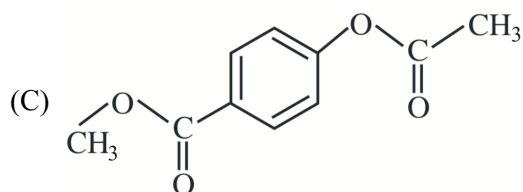
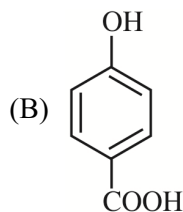
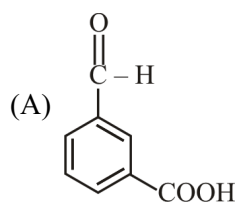
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. For the given conversion correct option among the following is



- (A) Maximum number of RMgX molecule that can be consumed by one molecule of A is 6
- (B) Total number of CO_2 molecule release on heating one molecule of C is 2
- (C) Final product D is a ketone
- (D) Final product D is an aldehyde and can consume only 1 molecule of RMgX

2. Which of the following has benzene ring where both the groups are deactivating in the nature?

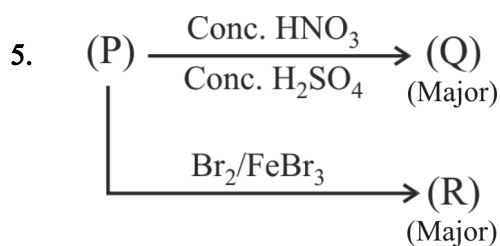


3. Among the following, which statement is not correct ?

- (A) When phenol is reacted with $\text{CHCl}_3 + \text{NaOH}$ then product is salicylaldehyde.
 (B) Ester is formed as major product when acetic acid treated with $\text{P}_2\text{O}_5/\Delta$
 (C) Fehling's solution is a good reagent to detect aromatic aldehydes
 (D) Both aldehyde and ketone can react with 2,4-dinitrophenylhydrazine reagent

4. $\text{CH}_3(\text{CH}_2)_{14}\text{COOH} \xrightarrow{\text{LiAlH}_4} \text{W} \xrightarrow{\text{HCl}} \text{X} \xrightarrow[2. \text{Ethylene oxide}]{1. \text{Mg/Et}_2\text{O}} \text{Y} \xrightarrow{\text{KMnO}_4/\text{H}^+} \text{Z}$
 Palmitic acid
 Choose correct option

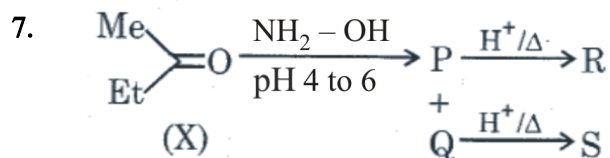
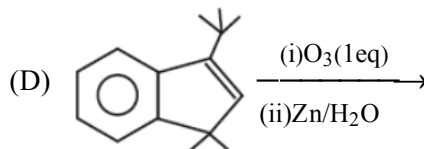
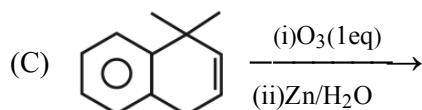
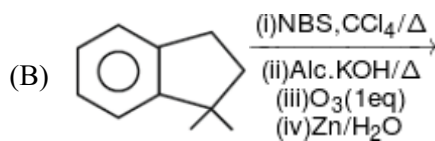
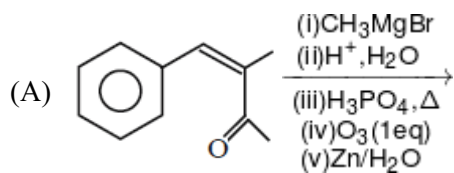
- (A) W is $\text{CH}_3(\text{CH}_2)_{14}\text{CH}_2\text{OH}$
 (B) X is $\text{CH}_3(\text{CH}_2)_{14}\text{CH}_2\text{Cl}$
 (C) Y is $\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$
 (D) Z is $\text{CH}_3(\text{CH}_2)_{16}\text{CHO}$



(P) was molecular formula $\text{C}_7\text{H}_6\text{O}_2$ which forms ' CO_2 ' gas with NaHCO_3 . Choose the incorrect option(s)?

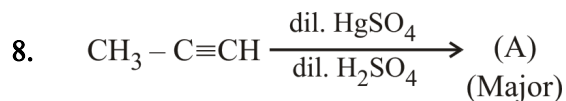
- (A) Acidic strength order : (Q) > (P) > (R)
 (B) (P) has IUPAC name p-Nitrobenzoic acid
 (C) (R) has common name o-Bromobenzoic acid
 (D) In formation of (Q) NO_2^+ is electrophilic while for (R) Br^+ is electrophile

6. In which of the following reaction, final major product gives intramolecular aldol condensation reaction ?



Which of the following option is/are correct?

- (A) P and Q are geometrical isomer
- (B) R and S are metamer
- (C) P to R is known as Beckmann rearrangement
- (D) X to P is known as Clemenson reduction



Compound (A) and PhCHO can not be differentiated by :

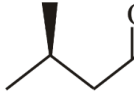
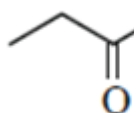
- (A) Fehling solution
- (B) Lucas test
- (C) 2,4-DNP test
- (D) Iodoform test

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

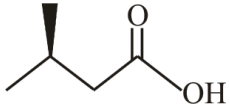
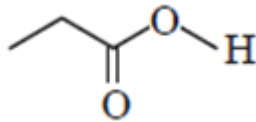
9. Answer the following by appropriately matching the lists based on the information given in the paragraph

List-I		List-II	
(I)	 $\xrightarrow[\text{(iii) Ag}_2\text{O}/\Delta \text{ (iv) HOH}]{\text{(i) SOCl}_2 \text{ (ii) CH}_2\text{N}_2}$	(P)	Substrate is unable to evolve CO ₂ on treatment with NaHCO ₃
(II)	 $\xrightarrow[\text{(ii) Br}_2/\text{CCl}_4]{\text{(i) AgOH}}$	(Q)	No dehydration but decarboxylation takes place in the reaction.
(III)	$\text{CH}_3 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{OH} \xrightarrow[\text{Catalytic amount}]{\text{P/Br}_2}$	(R)	Racemic mixture obtained in the final product
(IV)	$\text{H} - \overset{\text{O}}{\parallel} \text{C} - \overset{\text{O}}{\parallel} \text{C} - \text{H} \xrightarrow[\text{(iii) } \Delta]{\text{(i) OH}^- \text{ conc. (ii) H}^+}$	(S)	Number of C atom increases in the final product
		(T)	Reactant can undergo decarboxylation on heating
		(U)	Product obtained with lesser number of C atom as compare to reactant.

Which of the following option contains all correct combinations?

- (A) I → P; II → Q; III → T; IV → R
 (B) I → P; II → Q; III → R; IV → T
 (C) I → S; II → U; III → R; IV → P
 (D) I → S; II → U; III → P; IV → R

10. Answer the following by appropriately matching the lists based on the information given in the paragraph

List-I		List-II	
(I)	 $\xrightarrow[\text{(iii) Ag}_2\text{O}/\Delta \text{ (iv) HOH}]{\text{(i) SOCl}_2 \text{ (ii) CH}_2\text{N}_2}$	(P)	Substrate is unable to evolve CO ₂ on treatment with NaHCO ₃
(II)	 $\xrightarrow[\text{(ii) Br}_2/\text{CCl}_4]{\text{(i) AgOH}}$	(Q)	No dehydration but decarboxylation takes place in the reaction.
(III)	$\text{CH}_3 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{OH} \xrightarrow[\text{Catalytic amount}]{\text{P/Br}_2}$	(R)	Racemic mixture obtained in the final product
(IV)	$\text{H} - \overset{\text{O}}{\parallel} \text{C} - \overset{\text{O}}{\parallel} \text{C} - \text{H} \xrightarrow[\text{(ii) H}^+ \text{ (iii) } \Delta]{\text{(i) OH}^- \text{ conc.}}$	(S)	Number of C atom increases in the final product
		(T)	Reactant can undergo decarboxylation on heating
		(U)	Product obtained with lesser number of C atom as compare to reactant.

Which of the following is true. (about the substrate / reaction given above)?

- (A) I → S; II → U; III → R; IV → P, R
 (B) I → R, S; II → Q, T; III → U; IV → P, U
 (C) I → S; II → Q, U; III → R; IV → P, S
 (D) I → R; II → T; III → U; IV → S

11.

List-I		List-II (A and B is differentiated by)	
(I)	$\text{Ph} - \text{CH} = \text{O} \xrightarrow[(2) \text{H}^+]{(1) \text{Conc. } \text{OH}^-} \text{A} + \text{B}$	(P)	Tollen's test
(II)	$\text{CH}_3 - \text{C} \equiv \text{CH} \xrightarrow[\text{dil. H}_2\text{SO}_4]{\text{Hg}^{+2}} \text{A}$ $\text{CH}_3 - \text{C} \equiv \text{CH} \xrightarrow[(2) \text{H}_2\text{O}_2, \text{HO}^-]{(1) \text{BH}_3, \text{THF}} \text{B}$	(Q)	Fehling test
(III)	$\text{Ph} - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{OH} \xrightarrow{\Delta} (\text{A})$ $\text{CH}_3 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \underset{\text{CH}_3}{\text{CH}} - \overset{\text{O}}{\parallel} \text{C} - \text{OH} \xrightarrow{\Delta} (\text{B})$	(R)	NaHCO_3
(IV)	$\text{C}_6\text{H}_5\text{OCH(CH}_3)_2 \xrightarrow{\text{H}^+/\text{H}_2\text{O}} \text{A} + \text{B}$	(S)	Iodoform test
		(T)	FeCl_3
		(U)	2,4-DNP

The incorrect option is :

- (A) I $\rightarrow$ R; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ S
 (B) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ S
 (C) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ S; IV $\rightarrow$ T
 (D) I $\rightarrow$ R; II $\rightarrow$ T; III $\rightarrow$ S; IV $\rightarrow$ U

12.

List-I		List-II (A and B is differentiated by)	
(I)	$\text{Ph} - \text{CH} = \text{O} \xrightarrow[(2) \text{H}^+]{(1) \text{Conc. } \text{OH}^-} \text{A} + \text{B}$	(P)	Tollen's test
(II)	$\text{CH}_3 - \text{C} \equiv \text{CH} \xrightarrow[\text{dil. H}_2\text{SO}_4]{\text{Hg}^{+2}} \text{A}$ $\text{CH}_3 - \text{C} \equiv \text{CH} \xrightarrow[(2) \text{H}_2\text{O}_2, \text{HO}^-]{(1) \text{BH}_3, \text{THF}} \text{B}$	(Q)	Fehling test
(III)	$\text{Ph} - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{OH} \xrightarrow{\Delta} (\text{A})$ $\text{CH}_3 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \underset{\text{CH}_3}{\text{CH}} - \overset{\text{O}}{\parallel} \text{C} - \text{OH} \xrightarrow{\Delta} (\text{B})$	(R)	NaHCO_3
(IV)	$\text{C}_6\text{H}_5 - \text{O} - \text{C(CH}_3)_2 \xrightarrow{\text{H}^+ / \text{H}_2\text{O}} \text{A} + \text{B}$	(S)	Iodoform test
		(T)	FeCl_3
		(U)	2,4-DNP

The correct option is :

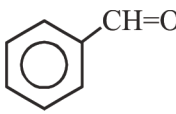
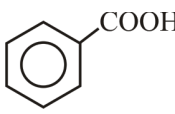
- (A) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ S; IV $\rightarrow$ Q
 (B) I $\rightarrow$ R; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ T
 (C) I $\rightarrow$ R; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ Q
 (D) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ S

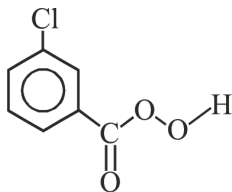
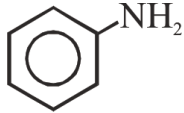
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

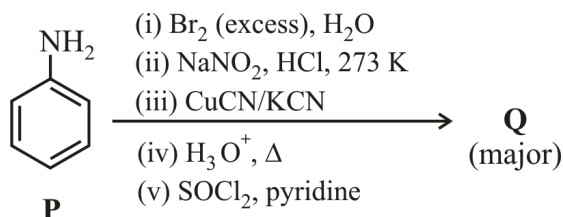
1. Number of reagents by which  can be converted in to  followed by acidification.

- (i) $[\text{Aq.}(\text{NH}_3)_2]\text{OH}$ (ii) 
- (iii) $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+, \Delta$ (iv) $\text{Conc. } ^-\text{OH}$
- (v) KMnO_4, Δ (vi) $\text{CuSO}_4 + \text{NH}_4\text{OH}$
- (vii) Zn-Hg/HCl (viii) Red P / HI
- (ix) Alcoholic KCN (x) 

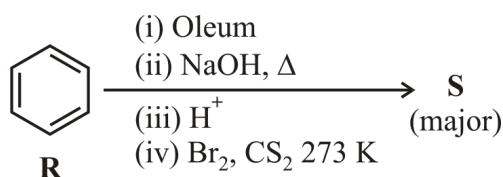
2. An organic compound ($\text{C}_8\text{H}_{10}\text{O}_2$) rotates plane-polarized light. It produces violet color with neutral FeCl_3 solution. What is the total number of all the possible isomers (including stereoisomers) for this compound?

3. Scheme 1 and 2 describe the conversion of P to Q and R to S, respectively. Scheme 3 describes the synthesis of T from Q and S. The total number of Br atoms in a molecule of T is _____

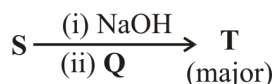
Scheme 1 :



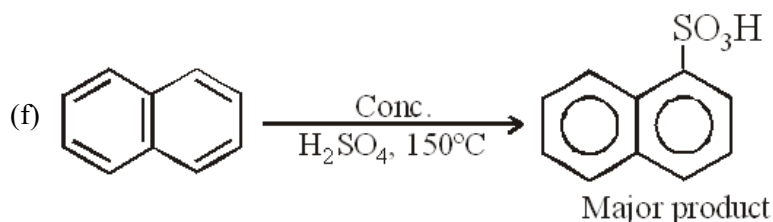
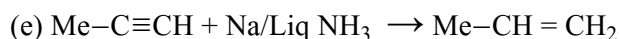
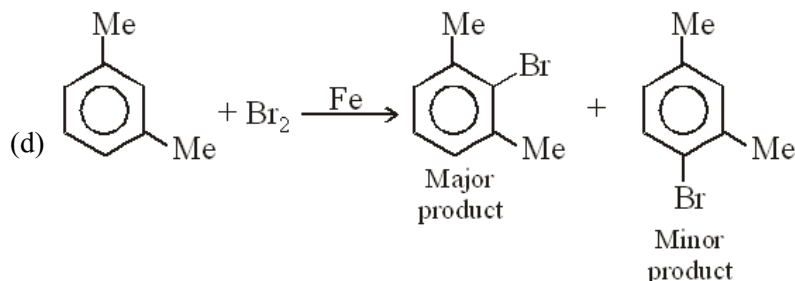
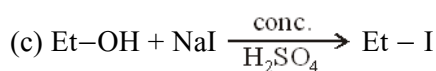
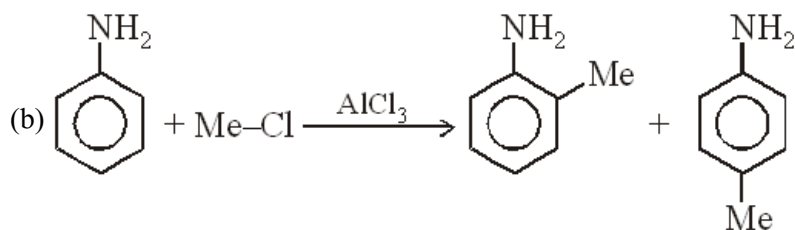
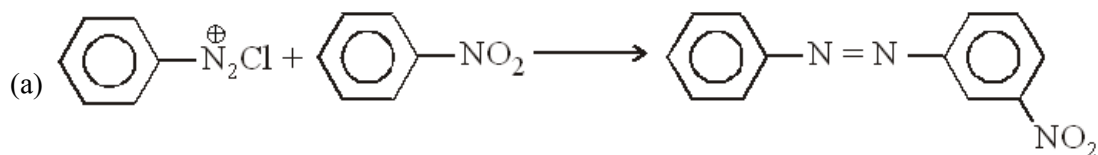
Scheme 2 :



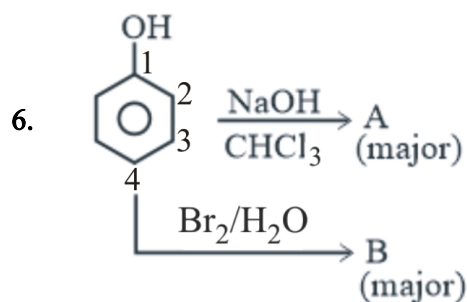
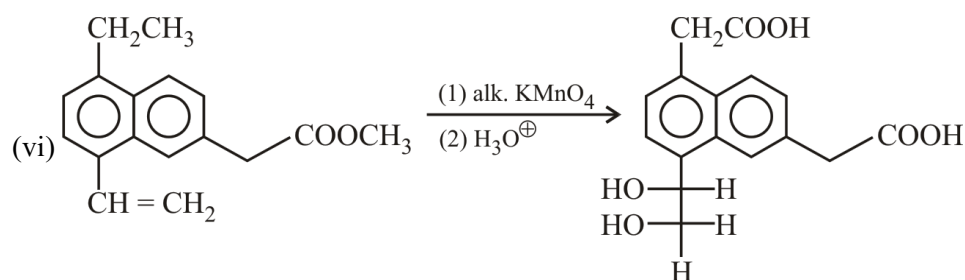
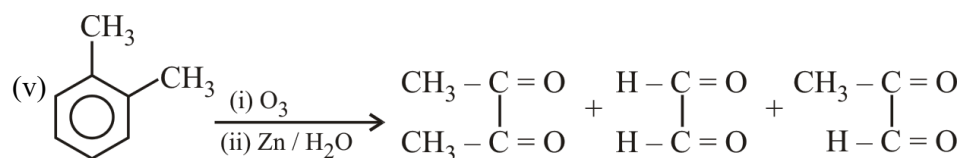
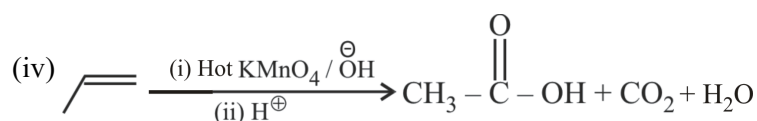
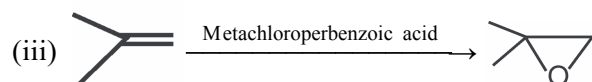
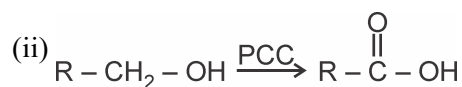
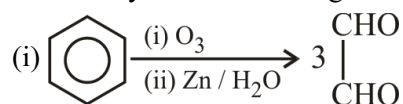
Scheme 3 :



4. Identify total number of reactions incorrectly match with its product ?



5. How many of the following are correctly matched with their product?



Position of CHO group in A is x, no. of Br atoms in B is y then $x + y = ?$

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. If $x\cos\alpha + y\sin\alpha = x\cos\beta + y\sin\beta = 2a$ and $2\sin\left(\frac{\alpha}{2}\right)\sin\left(\frac{\beta}{2}\right) = 1$ then
 - (A) $y^2 = 4a(a - x)$
 - (B) $\cos\alpha + \cos\beta = \cos\alpha \cos\beta$
 - (C) $\cos\alpha \cdot \cos\beta = \frac{4a^2 + y^2}{x^2 + y^2}$
 - (D) $(\cos\alpha + \cos\beta) = \frac{4ax}{x^2 + y^2}$
2. If $\cos\beta$ is geometric mean between $\sin\alpha$ and $\cos\alpha$, where $0 < \alpha, \beta < \frac{\pi}{2}$, then $\cos 2\beta =$
 - (A) $-2\sin^2\left(\frac{\pi}{4} - \alpha\right)$
 - (B) $-2\cos^2\left(\frac{\pi}{4} + \alpha\right)$
 - (C) $2\sin^2\left(\frac{\pi}{4} + \alpha\right)$
 - (D) $2\cos^2\left(\frac{\pi}{4} - \alpha\right)$

3. If $b > a > 0$, then the expression E given by

$$E = \frac{1}{\sqrt{b-a}} \frac{\sqrt{\frac{b-a}{a}} \sin x}{\sqrt{\left[1 + \left(\sqrt{\frac{b-a}{a}} \sin x\right)^2\right]}} \cdot \sqrt{a + b \tan^2 x} \text{ must be equal to}$$

- (A) $\tan x$
 (B) $|\tan x|$
 (C) $\frac{\sin x}{|\cos x|}$
 (D) unity if $x = \frac{11\pi}{4}$
4. If $\sin(x + 20^\circ) = 2 \sin x \cos 40^\circ$, where $x \in (0, \pi/2)$, then which of the following hold(s) good

- (A) $\cos 2x = \frac{1}{2}$
 (B) $\operatorname{cosec} 4x = 2$
 (C) $\sec \frac{x}{2} = \sqrt{6} - \sqrt{2}$
 (D) $\tan \frac{x}{2} = (2 - \sqrt{3})$

5. If $\cos \alpha + \cos \beta = a$,
 $2(\cos^2 \alpha + \cos^2 \beta) = b + 2$,
 $\cos 3\alpha + \cos 3\beta = c$, then which of the following options is/are correct ?

- (A) $\cos \alpha \cos \beta = \frac{a^2}{2} - \left(\frac{b+2}{4}\right)$
 (B) $2a^3 + c = 3a(1 + b)$
 (C) $a + b + c = 3abc$
 (D) $2(1 + b) = \frac{\sin 3\alpha}{\sin \alpha} + \frac{\sin 3\beta}{\sin \beta}$, for $\alpha, \beta \neq n\pi, n \in \mathbb{I}$

6. A given line L_1 intersects x and y axes at P and Q respectively. A variable line L_2 , perpendicular to L_1 , cuts x and y axes at R and S respectively. The locus of the point of intersection of the lines PS and QR is a curve passing through the

- (A) Origin
 (B) Point P
 (C) Point Q
 (D) Point R

7. If A(3, 4) and B(-5, -2) are the extremities of the base of an isosceles triangle ABC with $\tan C = 2$, then point C can be
- (A) $\left(\frac{3\sqrt{5}-1}{2}, -(1+2\sqrt{5}) \right)$
- (B) $\left(-\frac{(3\sqrt{5}+5)}{2}, 3+2\sqrt{5} \right)$
- (C) $\left(\frac{(3\sqrt{5}+5)}{2}, 3-2\sqrt{5} \right)$
- (D) $\left(\frac{(3\sqrt{5}+1)}{2}, -(1+2\sqrt{5}) \right)$
8. Two fixed points P and Q are 4 units apart and are on same side of a variable line L. Let PM and QN are perpendicular distance of P and Q from line L satisfying equation $PM + 3QN = 4$, then line L always touches a circle C then which among the following option(s) is/are correct for C ?
- (A) the centre of circle C lies on line PQ
- (B) the radius of circle is 1
- (C) the radius of circle is 2
- (D) the center of the circle C lies on perpendicular bisector of PQ

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. Let $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ be three distinct points lying on circle $S : x^2 + y^2 = 1$ such that

$$x_1x_2 + y_1y_2 + x_2x_3 + y_2y_3 + x_3x_1 + y_3y_1 = -\frac{3}{2}$$

List-I		List-II	
(I)	Let P be any arbitrary point lying on S, then $(PA)^2 + (PB)^2 + (PC)^2 =$	(P)	3
(II)	Let the perpendicular dropped from point 'A' to BC meets S at Q and $\angle OBQ = \frac{\pi}{k}$, where 'O' is origin, then $k =$	(Q)	4
(III)	Let R be the point lying on line $x + y = 2$, at the minimum distance from S and the square of maximum distance of R from S is $a + b\sqrt{b}$, then $a + b = \dots$ (Given a and b are distinct natural number)	(R)	6
(IV)	Let I and G represent incentre and centroid of $\triangle ABC$ respectively, then $IA + IB + IC + GA + GB + GC =$	(S)	5
		(T)	2
		(U)	7

Which of the following is the only correct combination :

- (A) $I \rightarrow R; II \rightarrow P; III \rightarrow S$
 (B) $I \rightarrow R; II \rightarrow P; IV \rightarrow T$
 (C) $I \rightarrow R; III \rightarrow U; IV \rightarrow R$
 (D) $II \rightarrow P; III \rightarrow T; IV \rightarrow R$

10. Let $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ be three distinct points lying on circle $S : x^2 + y^2 = 1$ such that

$$x_1x_2 + y_1y_2 + x_2x_3 + y_2y_3 + x_3x_1 + y_3y_1 = -\frac{3}{2}$$

List-I		List-II	
(I)	Let P be any arbitrary point lying on S, then $(PA)^2 + (PB)^2 + (PC)^2 =$	(P)	3
(II)	Let the perpendicular dropped from point 'A' to BC meets S at Q and $\angle OBQ = \frac{\pi}{k}$, where 'O' is origin, then k =	(Q)	4
(III)	Let R be the point lying on line $x + y = 2$, at the minimum distance from S and the square of maximum distance of R from S is $a + b\sqrt{b}$, then $a + b = \dots$ (Given a and b are distinct natural number)	(R)	6
(IV)	Let I and G represent incentre and centroid of ΔABC respectively, then $IA + IB + IC + GA + GB + GC =$	(S)	5
		(T)	2
		(U)	7

Which of the following is the only incorrect combination :

- (A) $I \rightarrow R; II \rightarrow P$
 (B) $III \rightarrow S; IV \rightarrow R$
 (C) $I \rightarrow R; III \rightarrow S$
 (D) $II \rightarrow R; IV \rightarrow T$

11. Match the set of values of x in List-II which satisfy the equations in List-I.

Let $D(0, \sqrt{3})$, $E(1, 0)$, $F(-1, 0)$ be the feet of perpendiculars dropped from vertices to the opposite sides BC, CA, AB respectively of triangle ABC.

List-I		List-II	
(I)	The ratio of the inradius of $\triangle ABC$ to the inradius of $\triangle DEF$ is	(P)	2
(II)	Let 'H' be the orthocentre of $\triangle ABC$, then the greatest integer which is less than or equal to square of the length AH is	(Q)	3
(III)	The square of the sum of ordinates of points A, B and C is	(R)	4
(IV)	The length of side AB of $\triangle ABC$ is	(S)	5
		(T)	1
		(U)	6

Which of the following is the only correct combination :

- (A) $I \rightarrow P; II \rightarrow S$
 (B) $II \rightarrow S; III \rightarrow T$
 (C) $III \rightarrow T; IV \rightarrow R$
 (D) $I \rightarrow P; IV \rightarrow U$

12. Match the set of values of x in List-II which satisfy the equations in List-I.

Let $D(0, \sqrt{3})$, $E(1, 0)$, $F(-1, 0)$ be the feet of perpendiculars dropped from vertices to the opposite sides BC, CA, AB respectively of triangle ABC.

List-I		List-II	
(I)	The ratio of the inradius of $\triangle ABC$ to the inradius of $\triangle DEF$ is	(P)	2
(II)	Let 'H' be the orthocentre of $\triangle ABC$, then the greatest integer which is less than or equal to square of the length AH is	(Q)	3
(III)	The square of the sum of ordinates of points A, B and C is	(R)	4
(IV)	The length of side AB of $\triangle ABC$ is	(S)	5
		(T)	1
		(U)	6

Which of the following is the only incorrect combination :

- (A) $I \rightarrow P; II \rightarrow S$
 (B) $III \rightarrow Q; IV \rightarrow R$
 (C) $I \rightarrow S; III \rightarrow P$
 (D) $II \rightarrow S; IV \rightarrow R$

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let x be in radians with $0 < x < \frac{\pi}{2}$. If $\sin(2\sin x) = \cos(2\cos x)$; then the value of $(\tan x + \cot x)$ can be written as $\pm \left(\frac{a}{\pi^c - b} \right)$ where $a, b, c \in \mathbb{N}$. Then the value of $\left(\frac{a+b+c}{25} \right)$ is
2. AB is a line segment of length 48cm and C is its mid-point. If three semicircles are drawn at AB, AC and CB using as diameters, then radius of the circle inscribed in the space enclosed by three semicircles is
3. If $P(x_0, y_0)$ be the coordinates of points on curve $5x^2 - 6xy + 5y^2 - 4 = 0$. Which is nearest to origin and $d(x_0, y_0)$ denotes its distance from origin then the value of $\frac{|x_0| + |y_0|}{(d(x_0, y_0))^2}$ is equal to
4. Let L_1, L_2 and L_3 be the length of tangents drawn from a point P to the circles $x^2 + y^2 = 4$, $x^2 + y^2 - 4x = 0$ and $x^2 + y^2 - 4y = 0$ respectively. If $L_1^4 = L_2^2 L_3^2 + 16$, then locus of P are the curve C_1 (a straight line) and C_2 (a circle). Let the circumcentre of the triangle formed by C_1 and other two lines which are inclined at an angle of 45° with C_1 and are tangents to C_2 is (α, β) then $(\alpha + 2\beta)$ equals
5. Let r_1 and r_2 be the radii of the largest and smallest circles, respectively, which pass through the point $(-4, 1)$ and having their centres on the circumference of the circle $x^2 + y^2 + 2x + 4y - 4 = 0$. If $\frac{r_1}{r_2} = a + b\sqrt{2}$ then $a + b$ is equal to :
6. A circle passes through the vertex C of a rectangle ABCD and touches its sides AB and AD at M and N, respectively. If the distance from C to the line-segment MN is equal to 5 units, then $\frac{1}{5}$ (area of the rectangle ABCD) =

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work

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